Linear Algebra Concepts

1. Norms & Inner Products

Definition of Norm: A norm is a function that assigns a strictly positive length or size to each vector in a vector space. In machine learning, vectors represent real-world objects, and their norms help quantify magnitude.

Length: The length of a vector is calculated through analytical geometry construction.

Distance: Measured as the geometric gap between two vectors. It is a critical metric for similarity — vectors that are close in distance are expected to yield similar outputs.

Dot Product & Orthogonality: Two vectors are orthogonal if their dot product is zero. Geometrically, this means they are perpendicular to each other.

2. Orthonormality

Unit Vectors: These are vectors with a length (norm) of 1.

Orthonormal Basis: A set of vectors forms an orthonormal basis if: 1. Every vector in the set is a unit vector. 2. Every pair of distinct vectors in the set is orthogonal. 3. The set is linearly independent and spans the space.

Standard Basis: In R^3, the standard orthonormal basis is i, j, k.

3. Eigenvalues & Eigenvectors

Core Definition: For a square matrix A, a scalar λ is an eigenvalue and a non-zero vector v is an eigenvector if $A v = \lambda v$. This means the transformation only scales the vector without changing its direction.

Characteristic Equation: Eigenvalues are found by solving $\det(A - \lambda I) = 0$.

For a 2x2 matrix: $\lambda^2 - (\text{Trace})\lambda + (\text{Determinant}) = 0$.

For a 3x3 matrix: $\lambda^3 - (\text{Trace})\lambda^2 + (\text{Sum of principal minors})\lambda - (\text{Determinant}) = 0$.

4. Orthogonal Matrices

A matrix Q is orthogonal if Q^T Q = I. This implies that the columns of Q are orthonormal.

Inverse Property: The inverse of an orthogonal matrix equals its transpose (Q^{-1} = Q^T).

Application: In the Least Squares Method, Q Q^T is used for orthogonal projection.

5. Diagonalization

A square matrix A is diagonalizable if A = P D P^{-1}, where P contains eigenvectors and D is a diagonal matrix of eigenvalues.

Requirement: A matrix is diagonalizable if and only if its eigenvectors form a basis for the space.

Why Diagonalize?
• Determinant: Product of diagonal entries.
• Powers: Raise each diagonal entry to the power n.
• Inverse: Reciprocal of each non-zero diagonal entry.